Departement Informatika
Department of Informatics

Assignment 02

Evaluation of Deep Learning Models on Image's Datasets

UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

University of Pretoria

Neetu Ramsaroop

□ **Length (5 – 10 pages). Excluding the cover page and reference page**

□ **Type: PDF file only. The page must include the student number and the student's proper name**

Submission Date: 06 October 2025 (23:59)

Submission Method: Online only

□ **Context:** You are provided with a dataset of screenshots from phishing and genuine websites. The goal is to explore unsupervised learning techniques using deep learning models to discover hidden patterns, anomalies, and relationships in visual data without using labels during training.

□ Learning Objectives:

By completing all tasks in this assignment, you will:

- Learn preprocessing and data cleaning for deep learning.
- Use CNN feature extraction and transfer learning to generate embeddings.
- Apply dimensionality reduction for visualization and interpretation.
- Perform clustering (k-means) to identify hidden groups or subgroups.
- Train autoencoders for anomaly detection.
- Perform association rule mining to discover frequent patterns.
- Integrate multiple unsupervised methods and critically analyze results.

Overall Skills Developed: Data preprocessing, feature extraction, visualization, clustering, anomaly detection, pattern discovery, and critical reflection on model performance and limitations.

□ Assignment Tasks:

Task 1 – Data Preprocessing

1.1 Load the dataset for **phishing** and **genuine** screenshots into the Jupyter Lab environment.

1.2 Pre-process the images by:

- Resizing all screenshots to a fixed size suitable for CNN input (e.g., 224×224).
- Normalizing pixel values to [0,1].
- Converting images to RGB if required.
- Apply optional data augmentation to increase dataset diversity and reduce overfitting:

  - Rotation

- Horizontal/vertical flipping
- Brightness adjustment
- Width/height shifts

- Optional masking or region-of-interest focus to remove irrelevant parts (like browser UI).

1.3 Justify why pre-processing is necessary for deep learning models and unsupervised analysis.

Task 2 – Feature Extraction Using Pretrained CNN

2.1 Use a pretrained CNN (e.g., ResNet50, MobileNetV2) as a feature extractor to generate embeddings for all preprocessed screenshots.

2.2 Explain why deep CNN embeddings are more suitable for unsupervised learning than raw pixel values.

2.3 Discuss what the embeddings represent (e.g., patterns, layout structures, color distribution).

Task 3 – Dimensionality Reduction

3.1 Apply t-SNE, UMAP, or PCA to reduce the feature embeddings to 2D or 3D.

3.2 Visualize the reduced embeddings in a scatter plot.

3.3 Analyze whether phishing and genuine screenshots form visually distinct clusters.

3.4 Justify the importance of dimensionality reduction for interpreting high-dimensional deep features.

Task 4 - K-Means Clustering

4.1 Perform k-means clustering on the feature embeddings with k=2 (phishing vs. genuine) and k>2 (exploring phishing subgroups).

4.2 Visualize clusters on the 2D embeddings from Task 3.

4.3 Evaluate cluster quality using Adjusted Rand Index (ARI) or Normalized Mutual Information (NMI).

4.4 Discuss the advantages and limitations of k-means for this dataset.

Task 5 - Autoencoder for Anomaly Detection

5.1 Build and train a convolutional autoencoder using only genuine website screenshots.

5.2 Compute reconstruction errors for both genuine and phishing images.

5.3 Explain why phishing images are expected to have higher reconstruction errors and how this can be used for anomaly detection.

Task 6 - Association Rule Mining

6.1 Convert screenshot features into binary/categorical data (e.g., presence of logos, login forms, color patterns).

6.2 Apply association rule mining (e.g., Apriori) to discover common patterns in phishing sites.

6.3 Explain how association rules can help identify frequent phishing strategies.

Task 7 - Interpretation and Reflection

7.1 Analyze the results of preprocessing, dimensionality reduction, clustering, autoencoder, and association rules.

7.2 Discuss insights about phishing website patterns and subgroups.

7.3 Reflect on how preprocessing affected model performance and clustering quality.

7.4 Suggest improvements or hybrid approaches for real-world unsupervised phishing detection.

Submission Instructions

Each student must submit the following:

- Report (PDF or Word). Your report must include:
  - Introduction and problem overview
  - Data Preparation and Preprocessing (Task 1)
  - Feature Extraction Using Pretrained CNN (Task 2)
  - Dimensionality Reduction (Task 3)
  - K-Means Clustering (Task 4)
  - Autoencoder for Anomaly Detection (Task 5)
  - Association Rule Mining (Task 6)
  - Interpretation and Reflection (Task 7)
- Preprocessed Dataset (CSV)
  - Submit a .csv file of your final, preprocessed dataset (After you load, resize, normalize, and optionally augment your images, the processed dataset is ready to be saved).
  - File name: yourname_preprocessed_screenshots.csv
- Jupyter Notebook or Python Script
    - Submit your complete notebook or .py file showing all tasks
    - Make sure your code is clean, well-commented, and follows the task sequence.
    - Executable end-to-end.
    - Includes all preprocessing, CNN feature extraction, dimensionality reduction, clustering, autoencoder, and optional association rule steps.
    - Visualizations can appear in appendices in the PDF.

Rubric

	Score	Comment
Task 1: Data Preparation	[19 marks]	
1.1 Dataset Loading & Cleaning - Correct loading of phishing and genuine screenshots - Handle missing/corrupted images - Clear folder structure	2 2 1	○ Images correctly loaded, missing/corrupt files identified and handled
1.2 Preprocessing and Augmentation - Resize to CNN input size - Normalize pixel values [0,1] - RGB conversion - Augmentation/masking	3 3 2 2	○ All steps applied correctly; justification provided [masking is optional]
1.3 Summary and Justification - Explain necessity of preprocessing for deep learning and unsupervised analysis - Include brief dataset overview	2 2	○ Clear rationale and summary provided
Task 2: Feature Extraction	[10 Marks]	
2.1 CNN Feature Extraction - Correct implementation of pretrained CNN - Embeddings generated for all images	3 2	○ Correct embeddings produced and verified
2.2 Explanation and Interpretation - Why CNN embeddings are better than raw pixels - What embeddings represent (patterns, layout, color)	3 2	○ Insightful explanation linked to unsupervised tasks
Task 3: Dimensionality Reduction	[10 Marks]	
3.1 Method Application - Correct use of t-SNE, UMAP, or PCA - Reduction to 2D/3D	3 2	○ Technique applied correctly
3.2 Visualization and Analysis - Scatter plot of embeddings - Assess separability of phishing vs genuine	3 2	○ Clear plot with brief analysis
Task 4: K-Means Clustering	[14 Marks]	
4.1 Clustering - K-Means for k=2 & k>2 - Correct implementation	3 2	○ Clusters generated correctly

4.2 Evaluation - Metrics: ARI, NMI - Interpretation of results	 3 2	○ Metrics computed and insights provided
4.3 Visualization - Overlay clusters on dimensionality reduction plots	2	○ Visualization clear, clusters labeled
4.4 Discussion - Advantages & limitations of K-Means	2	○ Thoughtful discussion included
Task 5: Autoencoder for Anomaly Detection	[14 Marks]	
5.1 Architecture and Training - Correct autoencoder architecture - Trained on genuine screenshots	 3 2	○ Autoencoder built and trained correctly
5.2 Reconstruction Error Analysis - Compute errors for phishing and genuine - Histogram/visualization	 3 1	○ Errors computed and plotted
5.3 Interpretation - Justify why phishing has higher error - Explain anomaly detection use	 3 2	○ Clear explanation provided
Task 6: Association Rule Mining	[7 Marks]	
6.1 Feature Conversion - Convert image features to binary/categorical	2	○ Properly converted
6.2 Rule Mining - Apply Apriori/association rules	2	○ Correct implementation
6.3 Interpretation - Insights about phishing strategies	3	○ Clear interpretation provided
Task 7: Interpretation and Reflection	[4 Marks]	
7.1 Results Analysis - Summarize results from preprocessing, embeddings, clustering, autoencoder	2	○ Concise and clear analysis
7.2 Insights and Recommendations - Discuss phishing patterns, preprocessing impact, hybrid methods	2	○ Thoughtful insights & practical recommendations
Presentation & Report Quality	[2 Marks]	
Clarity, conciseness, referencing appendices, captions. Includes Introduction and Conclusion	2	○ Report well-structured, appendices referenced
Total	**80**	

Suggested Report Structure (10 Pages)

Cover Page (1 page)

Title: e.g., Evaluation of Deep Learning Models on Image's Datasets

Name, Student ID, Course, Date

<u>Assignment 2</u>

1. Introduction [0.5 – 1 page]

1.1 Describe the phishing detection problem and its significance.
1.2 State the objectives of the assignment.
1.3 Briefly describe the dataset (phishing vs. genuine screenshots).

2. Data Preparation and Preprocessing (Task 1) [1 – 1.5 pages]

2.1 Summarize dataset loading and preprocessing steps:
- Resizing to CNN input size (e.g., 224×224)
- Normalization of pixel values [0,1]
- RGB conversion
- Augmentation
- Optional: masking

2.2 Justify why preprocessing is necessary for deep learning and unsupervised analysis.

- **Reference figures in Appendix A for sample images.**

3. Feature Extraction Using Pretrained CNN (Task 2) [1 page]

3.1 Specify the CNN model used (e.g., ResNet50, MobileNetV2).
3.2 Explain why embeddings are more suitable than raw pixels for unsupervised learning.
3.3 Discuss what the embeddings represent (patterns, layout, color distribution).

- **Reference figures in Appendix B for embedding visualizations.**

4. Dimensionality Reduction (Task 3) [1 page]

4.1 Describe the method used (t-SNE, UMAP, PCA).
4.2 Analyze whether phishing and genuine screenshots form visually distinct clusters.
4.3 Justify the importance of dimensionality reduction for high-dimensional features.

- **Reference figures in Appendix C for scatter plots.**

5. K-Means Clustering (Task 4) [1 – 1.5 pages]

5.1 Describe clustering experiments:
- k=2 (phishing vs. genuine)
- k>2 (exploring phishing subgroups)

5.2 Evaluate cluster quality using ARI and NMI metrics.

5.3 Discuss advantages and limitations of k-means for this dataset.

- **Reference figures in Appendix D for cluster visualizations.**

6. Autoencoder for Anomaly Detection (Task 5) [1 – 1.5 pages]

6.1 Describe convolutional autoencoder architecture and training (on genuine screenshots).
6.2 Compute and visualize reconstruction errors for phishing vs. genuine images.
6.3 Justify why phishing images have higher reconstruction errors.

- **Reference figures in Appendix E for reconstruction error histograms.**

7. Association Rule Mining (Optional, Task 6) [0.5 pages]

7.1 Convert screenshot features into binary/categorical data (logos, login forms, color patterns).

7.2 Apply Apriori or other rule mining techniques.

7.3 Discuss insights from association rules about phishing strategies.

- **Reference figures/tables in Appendix F for association rules.**

8. Interpretation and Reflection (Task 7) [1 page]

8.1 Analyze and summarize results from all previous tasks.

8.2 Discuss insights about phishing patterns, subgroups, and model performance.

8.3 Reflect on the effect of preprocessing on clustering and anomaly detection.

8.4 Suggest improvements or hybrid approaches for real-world phishing detection.

9. Visualizations

- All plots, charts, and diagrams should appear in appendices with captions.
- Within the main report, reference figures as: “See Appendix A for sample screenshots.”

Other aspects to include:

- Model Choice and Justification [0.5 page (integrated into Tasks 2–5 if needed)]
  - Explain why each model or method was chosen.
  - Include rationale for CNN embeddings, dimensionality reduction, clustering, and autoencoder.
- Evaluation Metrics and Interpretation [0.5 – 1 page (integrated with Tasks 4–5)]
  - Include quantitative results (ARI, NMI, reconstruction error).
  - Provide interpretation and implications for phishing detection.

- Key Findings and Recommendations [0.5 page]
  - Summarize major insights.
  - Provide actionable recommendations for deploying unsupervised phishing detection systems.

Appendices (Optional, not counted toward 10 pages)

- Appendix A: Sample preprocessed images (Task 1)
- Appendix B: CNN feature embeddings visualization (Task 2)
- Appendix C: Dimensionality reduction scatter plots (Task 3)
- Appendix D: K-means cluster visualizations (Task 4)
- Appendix E: Autoencoder reconstruction error histograms (Task 5)
- Appendix F: Association rule mining results (Task 6)
- Appendix G: Complete Jupyter Notebook or Python code (Tasks 1–6)

Note: Label each figure and table clearly, e.g., "Figure A1: Sample genuine and phishing screenshots after preprocessing."

Tips to Stay Within 10 Pages:

- Use figures and tables efficiently (combine multiple visuals into one if needed).
- Keep code snippets short in the main report; link or refer to full code in the appendix.
- Use concise bullet points or tables for explanations where appropriate.

Referenced Information. To be able to complete this assignment successfully, the student should reference the following sources, in the following order of priority:

Due Dates & Times. ASSIGNMENTS SUBMITTED LATE WILL NOT BE MARKED. NO EXCEPTIONS, NO EXCUSES.

Submission Instructions. This is an ***INDIVIDUAL ASSIGNMENT***. All assignments will be submitted to Turnitin to assess the originality or the similarity of assignment content. If duplication is found between different sets of project documentation, then the person involved will be penalized.

While AI tools like ChatGPT can assist you, it is crucial that your submissions reflect your own understanding and analysis. Copying or heavily relying on AI-generated content undermines your learning experience and academic integrity. The use of AI tools as a supplement to your own thinking will be penalized.

Assignment Layout and Appearance

- No handwritten work will be accepted.

 The manuscript should contain a title page (should be captivating), your name, student number, the course name and instructor's name. The manuscript should only contain an introduction, body, and conclusion.

 All graphs, images or diagrams should be clearly visible.

- Content on the graphs, images or diagrams should be clearly legible.
- Graphs, images or diagrams should be clearly labeled and entitled.

 Assignment Referencing.

- References are required for this assignment.
- If a student makes use of any sources, it will be required for students to reference the sources used.
- If a student makes use of direct quotations from any source, then it will be required that the student make use of Harvard referencing and appropriate citation mechanisms.

Penalties. Penalties will also apply to issues relating to page length, spelling, punctuation, grammar, and figure manipulations.